ASUS ROG Astral GeForce RTX 5080 16GB GDDR7 OC

Premium Card, Premium Price

₹1,69,950

The arrival of NVIDIA's Blackwell GPU lineup marks a notable step forward in graphics technology, and the GeForce RTX 5080 is intended as a carefully balanced option between raw power and accessibility. Priced at US$999, this mid-tier model provides a more attainable entry into the Blackwell family without sacrificing too much of the series' advancements. It carries 10,752 CUDA cores, which is roughly half the complement found in the higher-end RTX 5090, and it is built on the same TSMC 4N process for efficient power usage.

7.7

Performance 7.7
Features............... 7.5
Build Quality 8.3
Value for money... 7.5

ASUS has taken this foundation and applied its own enhancements with the ASUS ROG Astral GeForce RTX 5080 16GB GDDR7 OC Edition. Alongside the standard 16 GB of GDDR7 memory on a 256-bit bus, the ASUS variant offers an overclocked twist and a sleek dual-slot design. The card's 45.6 billion transistors on a 378 mm^2 die, combined with NVIDIA's DLSS4 upscaling technology, position it as a capable performer across intensive gaming, content creation, and even machine learning tasks. In this review, we will explore the ASUS ROG Astral edition's design, performance benchmarks, and how it fits into the current landscape of high-end graphics solutions.

SPECIFICATIONS

Built on the Blackwell GB203 GPU, the GeForce RTX 5080 features 84 active Streaming Multiprocessors, totalling 10,752 shader cores along with 112 ROPs and 336 texture units. The base clock is 2295 MHz, with a boost up to 2760 MHz. Then there's the fact that all 50-series cards carry 4th gen Ray Tracing Cores and 5th gen Tensor Cores for real-time ray tracing and AI workloads. The card's 16 GB of GDDR7 runs on a 256-bit interface, reaching 30 Gbps or cumulatively that's an effective bandwidth of 960 GB/s.

Memory technology advancements, including more efficient compression algorithms, enhance overall performance, minimising bottlenecks during high-resolution gaming sessions and resource-intensive workloads. The larger caches also contribute to smoother frame delivery, helping the RTX 5080 maintain reliable frame rates in modern titles. Cooling solutions must manage a Total Board Power of 360 Watts, although typical power demands are generally lower in most gaming scenarios.

While the Founders Edition model employs a dual-slot cooler design, aftermarket variants such as the ASUS ROG Astral GeForce RTX 5080 16GB GDDR7 OC Edition opt for more elaborate systems for further thermal control. We get to see four cooling fans on the Astral variant with three of them on the front and the fourth fan on the rear of the card for a push-pull combination. This does make the card quite beefy and ASUS has included a support peg in the packaging to prop the card up.

As a result, the RTX 5080 delivers a balanced combination of computational prowess, efficient memory usage, and feature-rich design that aims to appeal to a broad range of users. Its multifaceted capabilities are intended to serve gamers, creators, and researchers as well. Folks who'd want to experiment with DeepSeek might eye this card for its massive memory buffer and high core count.

PERFORMANCE

We have featured some of the older series cards along with most of the RTX 30 and RTX 40 series cards including the Super cards. A few AMD RX 7000 series cards were also thrown into the mix to see how good the cards perform against the competition. With barely 24 hours on hand, we've been putting the card through its paces and this review will be updated as and when we finish more benchmarks. Like always, we have a section for synthetic benchmarks